

Flow-Imbalance Long/Short Strategy (Perpetual Futures)

Data and signal construction

We build a daily long/short strategy on Binance perpetual futures using order-flow imbalance as the signal. For each symbol **i** and day **t**, we aggregate all hourly perp klines and compute:

Formula: $\text{flow_imbalance_1d}(i, t) = \frac{\sum_h \text{taker_buy_quote_volume}(i, t, h)}{\sum_h \text{quote_volume}(i, t, h)}$.

- Universe: Top 100 symbols by trailing 30-day average dollar volume, recomputed each day from `daily_returns.parquet`. This focuses on liquid contracts and is similar to the liquidity filter used in the extra-credit analysis. - Signal date: Day *T*. We use all hourly bars from 00:00:00 UTC to form `flow_imbalance_1d(i, T)`.

Trading timeline and implementation

The key implementation choice is when the signal is known relative to the traded return:

- The last hourly bar for day *T* (23:00-24:00 UTC) is available shortly after midnight at the start of day *T+1*. - We therefore trade using `ret_utc1` on day *T+1*, which is the 24-hour return from 01:00 UTC *T+1* to 01:00 UTC *T+2*. - This creates at least a one-hour buffer between signal availability and the start of the traded return and avoids any look-ahead bias.

If we instead tried to trade on returns that start before the final bar of day *T* is known (for example, using `ret_utc0` on the same day), the backtest would implicitly see into the future and produce unrealistically high Sharpes. The `run_q3.py` code enforces the *T* -> *T+1* mapping by building a panel with `signal_date = sd` and `trade_date = sd + 1`.

Portfolio construction and backtest

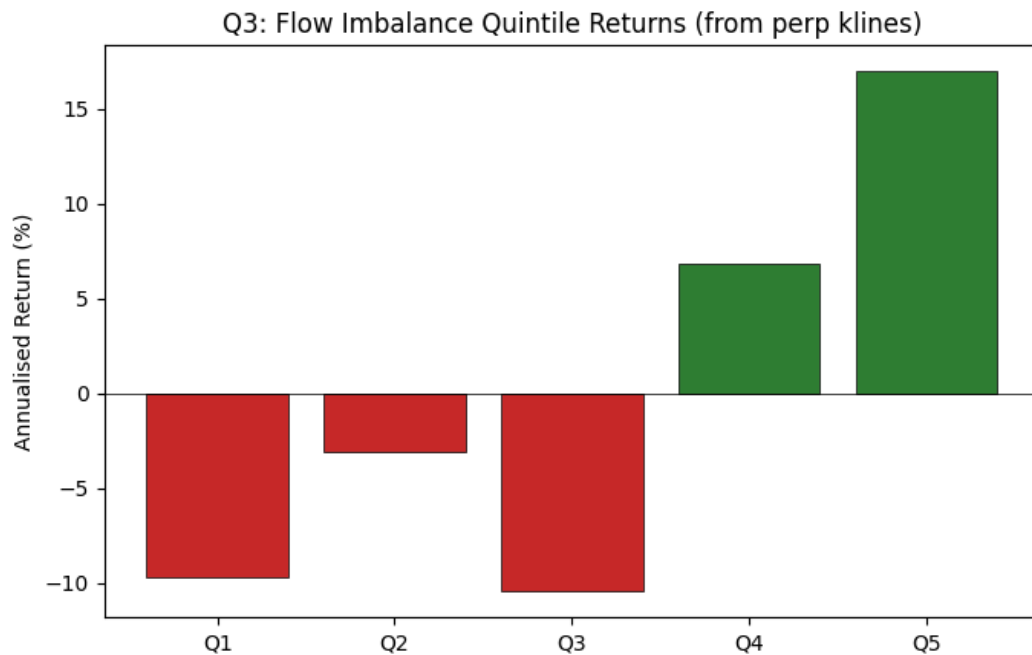
Each trading day from 2023-01-01 onwards:

- We keep only symbols in the top-100 liquidity universe on signal date *T*. - We perform a quintile sort on flow imbalance: - Q5: highest buying pressure (most aggressive taker buys). - Q1: lowest buying pressure. - Within each quintile, positions are equalweighted. - The long/short portfolio is long Q5, short Q1, rebalanced daily.

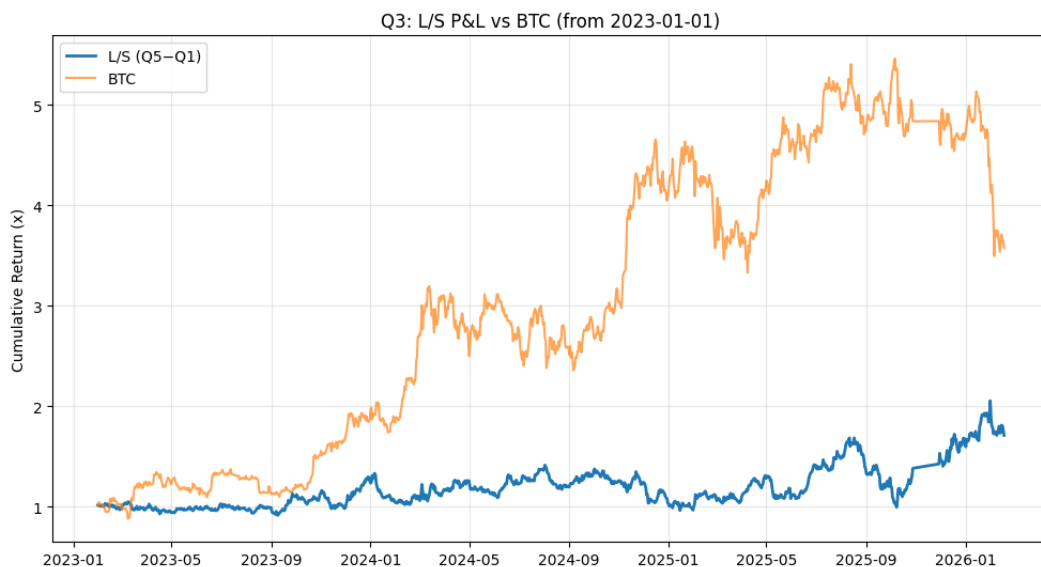
Over the full sample the strategy delivers:

- Long/short (Q5-Q1): - Sharpe: 0.64 - Annualised return: 26.6% - Annualised volatility: 41.3% - Cumulative return: 71.2% - Max drawdown: 40.9%

These statistics are read from Q3/results/q3_metrics.txt and match the base results reported in the extra-credit write-up. The quintile bar chart below shows a clear spread from deeply negative returns in Q1 to strongly positive returns in Q5, consistent with order flow containing predictive information for next-day returns.



The cumulative P&L; chart compares the long/short strategy to BTC:



BTC experiences a larger overall run-up and bigger swings, while the market-neutral long/short strategy compounds steadily with smaller drawdowns. This behaviour is typical for cross-sectional factor portfolios.

Factor exposure and BTC alpha

To understand how much of the performance is simply a disguised BTC bet, `run_q3.py` runs a daily regression of L/S returns on BTC returns:

Regression: $L/S_t = \alpha + \beta \cdot r_{BTC,t} + \text{error}_t$.

From `q3_metrics.txt`:

- Annualised alpha: 32.6% per year. - Beta to BTC: 0.11 (slightly short BTC overall). - t-stat of alpha: 1.37. - R^2 : 1.6%.

The negative beta confirms that the strategy is not merely long the market; if anything it slightly hedges BTC. The alpha estimate is economically large but only marginally statistically significant over the sample, which is consistent with a noisy, high-volatility environment.

The assignment does not provide Fama-French factor data, and standard FF5 + MOM factors (SMB, HML, RMW, CMA, MOM) are constructed from equity markets and carry no direct meaning in crypto asset markets. We therefore do not run a traditional FF5 regression. Instead, we use a crypto-native proxy factor regression regressing L/S returns on BTC, ETH, and an equal-weighted market return across the top-100 universe as the appropriate analogue for factor exposure analysis.

Crypto proxy factor regression (BTC, ETH, market)

As a robustness check we run a multi-factor regression of the L/S returns on BTC, ETH and an equal-weighted market return across the top-100 universe. The estimated daily alpha is about 0.00074 (roughly 27% annualised) with an R^2 around 4%, and small betas to all three factors (BTC 0.08, ETH 0.12, MktRF 0.04). This suggests that broad BTC/ETH/market moves explain only a small fraction of the strategy's variance and that most of the performance reflects stock-selection across names rather than simple crypto beta exposure.

Behaviour in the early2026 meltdown

To stress-test the strategy, we restrict the sample to January-March 2026, when the broader crypto market experienced a sharp drawdown. Over this window the long/short portfolio achieves:

- Sharpe: 0.55 - Annualised return: 33.3% - Annualised volatility: 61.1% - Cumulative return: 2.0% - Max drawdown: 16.7%

While BTC suffers a substantial sell-off in this period, the market-neutral portfolio remains slightly profitable with substantially smaller drawdowns, highlighting that flow-imbalance captures relative-value opportunities that are less sensitive to broad market direction.

Interpretation

Putting these pieces together:

- The timing of the signal ($T \rightarrow T+1$ with `ret_utc1`) ensures that the backtest is implementable and that strong performance is not an artefact of look-ahead. - The quintile pattern and the long/short metrics suggest that daily order-flow imbalance is a meaningful predictor of next-day cross-sectional returns in liquid crypto perps. - The strategy exhibits moderate standalone Sharpe with sizeable but manageable drawdowns, and it behaves relatively well during severe market stress, making it a plausible building block for a diversified quant portfolio.